# Iterative Linear Probing and Counterfactual Importance: Redundancy in Belief-State Representations

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## 1 Introduction

Previous probing experiments on transformers trained on Hidden Markov Model (HMM) data have established that the residual stream encodes a high-fidelity linear image of the Bayesian belief state (the posterior distribution over hidden states given the observed token history). However, initial ablation experiments—in which the three SVD directions of a single linear probe's coefficient matrix were mean-ablated one at a time—produced KL divergences only marginally larger than those from ablating random directions. This raised the possibility that the linearly decodable belief state is not the representation the model actually *uses* for prediction, or that the causal signal is distributed across more directions than the three extracted from a single probe.

These observations motivate two sharper questions: *how many independent copies* of the belief state does the network store in its residual stream, and *which copies matter* for the model's actual next-token predictions?

These questions bear on two distinct aspects of the representation:

1. **Redundancy.** A 64-dimensional residual stream need only devote 2 dimensions to represent the Mess3 belief state (which lives on a 2-simplex). If the network stores multiple independent copies, this tells us something about how superposition and redundancy interact in small transformers—and whether the network "wastes" capacity or exploits it for robustness.

2. **Causal relevance.** A subspace may be linearly decodable without being causally used by downstream layers. Counterfactual ablation (replacing a subspace's projection with its mean) tests whether the model's output *changes* when that information is erased, distinguishing representation from mere correlation.

This report describes the *iterative probing* algorithm and its application to a 4-layer, 64-dimensional, single-head transformer trained on the Mess3 process.

# 2 Model and Process

The Mess3 process (`src/hmm.py`, class `Mess3Proc`) is a 3-state, 3-token HMM. The model is a 4-layer HookedTransformer with $d_{\text{model}} = 64$, a single attention head ($d_{\text{head}} = 64$), MLP width 256, ReLU activations, and LayerNorm. It was trained on 10M tokens with context length $n_{\text{ctx}} = 10$. The geometric structure of belief-state representations in transformers trained on HMMs has been studied by Shai et al. [1]. The model directory is:

```
models/mess3/20260306_150920_L4_d64_H1_full_LN/
```

The belief state for Mess3 has 3 components summing to 1, so it lives on a 2-dimensional simplex embedded in $\mathbb{R}^3$. A linear probe mapping $\mathbb{R}^{64} \to \mathbb{R}^3$ therefore has a coefficient matrix of shape $3 \times 64$ whose row space is generically 2-dimensional (or at most 3, depending on how the intercept interacts with the sum-to-one constraint).

# 3 Algorithm

## 3.1 Iterative Probing

The core idea is to repeatedly extract and remove the best linear predictor of the belief state from the activations, counting how many times this can be done before the residual signal drops below a significance threshold.

For each layer $\ell$:

1. Let $X^{(0)} \in \mathbb{R}^{n \times 64}$ be the last-position residual-stream activations.

2. At iteration $k$:

   (a) Fit a linear probe: $\hat{y} = X^{(k)} W^\top + b$, where $W \in \mathbb{R}^{3 \times 64}$.

   (b) Evaluate test $R^2$. If $R^2 < \tau$ (default $\tau = 0.05$), stop.

(c) Compute the SVD of $W = U\Sigma V^\top$. Retain rows of $V^\top$ corresponding to singular values above a relative threshold ($\sigma_i/\sigma_0 > 10^{-3}$). Call the retained rows $V_k \in \mathbb{R}^{r_k \times 64}$, where $r_k$ is the effective rank.

(d) Project out: $X^{(k+1)} = X^{(k)} - X^{(k)} V_k^\top V_k$.

3. The number of iterations with $R^2 \geq \tau$ is the number of "independent copies."

This is implemented in `src/iterative_probe.py`, with the main entry points being `iterative_probe_single_l` and `iterative_probe_all_layers()`. The function `extract_subspace()` performs the SVD and rank thresholding; `project_out_subspace()` handles the orthogonal projection. The algorithm reuses `fit_probe_with_split()` from `src/linear_probe.py` at each iteration to ensure a consistent 80/20 train/test split.

**Why $R^2$ and not MSE?** MSE depends on the scale of the targets, while $R^2$ is normalised. Because the projected activations have shrinking variance with each iteration, MSE alone could be misleading; $R^2$ measures the fraction of target variance explained regardless of the activation scale.

### 3.2 Counterfactual Importance via Mean Ablation

For each layer $\ell$ and each iteration-$k$ subspace $V_k$:

1. Run the model cleanly to obtain reference log-probabilities $\log p_{\text{clean}}$ at the last token position.

2. Intervene on the residual stream at layer $\ell$ using a `run_with_hooks` forward hook: project the activations onto $V_k$, replace the projection with its batch mean, and add back the residual. Formally, if $a$ is the activation vector:

$$a' = a - V_k^\top V_k\, a + \mathbb{E}_{\text{batch}}[V_k^\top V_k\, a]. \tag{1}$$

3. Compute the KL divergence: $D_{\text{KL}}(p_{\text{clean}} \| p_{\text{ablated}})$, which measures how much the model's output distribution shifts when this subspace is erased.

4. Also compute the cross-entropy change $\Delta$CE as a secondary metric.

This is implemented in `src/run_iterative_probe.py`, functions `ablate_subspace()` and `compute_counterfac` The ablation generalises the single-direction ablation in `archive/old_src/ablation.py` to arbitrary-dimensional subspaces via the einsum-based projection `t.einsum("bsd,rd->bsr", actvs, subspace)`.

We additionally ablate each individual SVD direction within a subspace and the cumulative subspace (all iterations combined) to obtain finer-grained attribution.

## 4 Results

The experiment was run with 10,000 sequences of length 10, using the last token position. The algorithm was run for all 32 possible iterations (exhausting the full 64-dimensional space with rank-2 subspaces), with no $R^2$ stopping threshold.

**Note on position selection.** We probe only the last sequence position, which gives 10,000 samples. Using all positions would increase the sample count 10-fold to 100,000, potentially improving probe stability and revealing subtler structure at early positions where the belief state has not yet converged. However, last-position probing is sufficient for the present analysis and avoids conflating position-dependent effects (e.g., the belief state is less well-determined at early positions) with the redundancy structure we aim to measure.

## 4.1 Iterative Probing: Many Redundant Copies

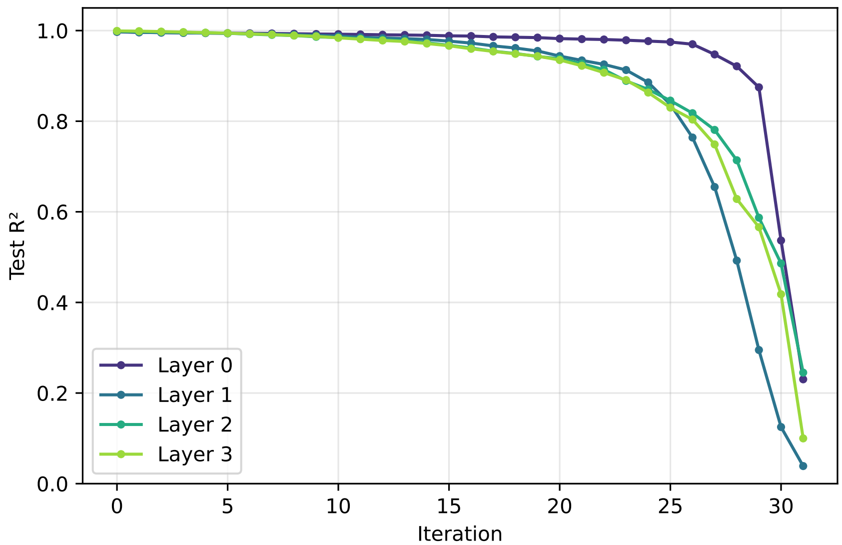


Figure 1: Test $R^2$ as a function of iteration for each layer across all 32 iterations. $R^2$ remains above 0.95 through ∼25 iterations before collapsing as the residual stream is exhausted.

Figure 1 shows the full trajectory of $R^2$ across all 32 iterations. Each iteration extracts a rank-2 subspace (consistent with the 2D simplex geometry of the Mess3 belief state), so 32 iterations exhaust all $32 \times 2 = 64$ dimensions of the residual stream.

The $R^2$ curve reveals three distinct regimes:

1. **Plateau** (iterations 0–20, 0–40 dims): $R^2$ remains above 0.93 at all layers, declining only gradually. The belief state is linearly decodable from at least 20 mutually orthogonal 2D subspaces.

2. **Transition** (iterations 20–28, 40–56 dims): $R^2$ begins to fall more steeply, particularly at later layers. By iteration 27 (54 dims removed), layer 3 drops to $R^2 \approx 0.69$.

3. **Collapse** (iterations 28–31, 56–64 dims): $R^2$ drops sharply as the remaining dimensions are insufficient to represent the belief state. The final iteration yields $R^2 \approx 0.19$–$0.24$, consistent with near-chance prediction from 2 remaining dimensions.

The degradation is not uniform across layers. Layer 0 retains the highest $R^2$ longest (0.93 at iteration 28), while layer 3 degrades fastest (0.62 at iteration 28). This suggests that the belief-state signal is spread more uniformly across dimensions at layer 0, while at later layers it is more concentrated in a smaller subspace.

Figure 2 shows the corresponding MSE growth. Later layers start with lower MSE (layer 3 begins at $1.3 \times 10^{-4}$, vs. $4.4 \times 10^{-4}$ for layer 0), reflecting a more precise belief-state encoding deeper in the network. The MSE growth is approximately linear through the first ∼40 dimensions removed, suggesting that the redundant copies contribute roughly equally to probe accuracy in this regime. Beyond 40 dimensions, the MSE curve steepens superlinearly, and by 64 dimensions (all projected out) the MSE reaches 0.13–0.16, comparable to predicting the marginal mean.

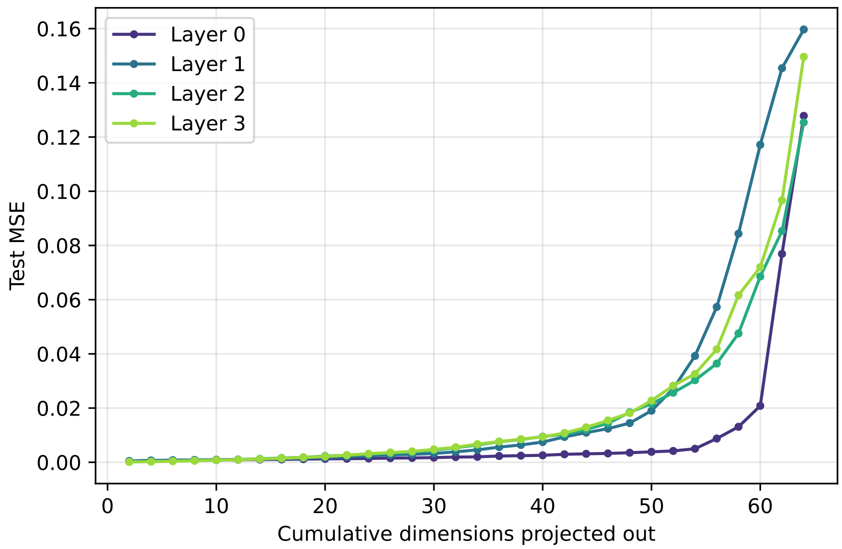


Figure 2: Test MSE vs. the number of dimensions projected out. MSE grows slowly through ∼40 dims, then accelerates as the remaining subspace can no longer support the belief state.

## 4.2 Counterfactual Importance: A Strong Hierarchy

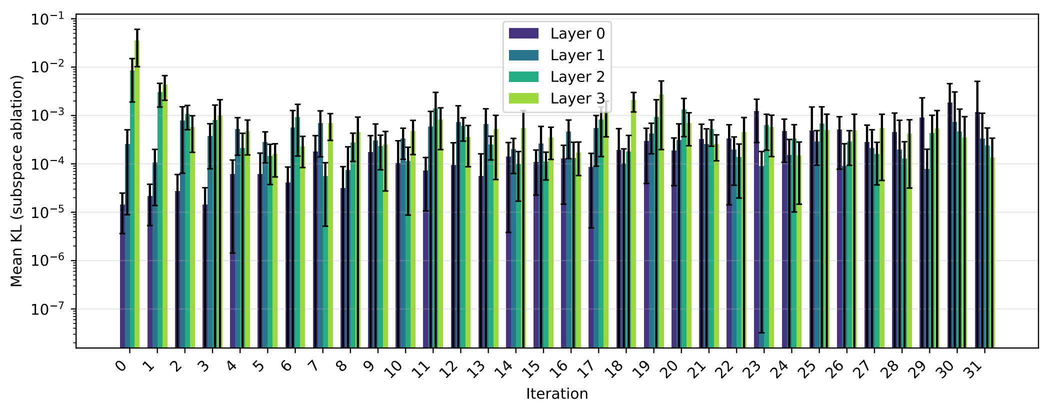


Figure 3: Mean KL divergence from ablating each iteration's subspace, broken down by layer. Note the log scale. Iteration 0 at layer 3 dominates by an order of magnitude.

Figure 3 reveals a strikingly different picture from the probing results. While all 32 subspaces at each layer are *decodable* (with $R^2 > 0.19$ even at the final iteration), they are far from equally *important*:

- **Layer 3, iteration 0** produces the largest KL shift by a wide margin: $\mathrm{KL} = 0.040 \pm 0.027$. The second iteration at layer 3 drops by 12× to $\mathrm{KL} = 0.003$. Subsequent iterations fluctuate around $10^{-4}$–$10^{-3}$, with no systematic pattern.
- **Layer 2** shows a similar but weaker pattern: iteration 0 gives $\mathrm{KL} = 0.009$, iteration 1 gives 0.002, then the signal drops to $\sim 10^{-4}$.

- **Layer 0** subspaces are almost entirely causally inert: individual iterations give KL $\sim 10^{-5}$–$10^{-4}$, though the later iterations (25–31) show slightly elevated KL ($\sim 10^{-3}$), likely because ablating the last few dimensions begins to disrupt the residual stream's overall structure regardless of belief-state content.

- The causal importance follows a clear **layer hierarchy**: layer 3 $\gg$ layer 2 $>$ layer 1 $\gg$ layer 0.

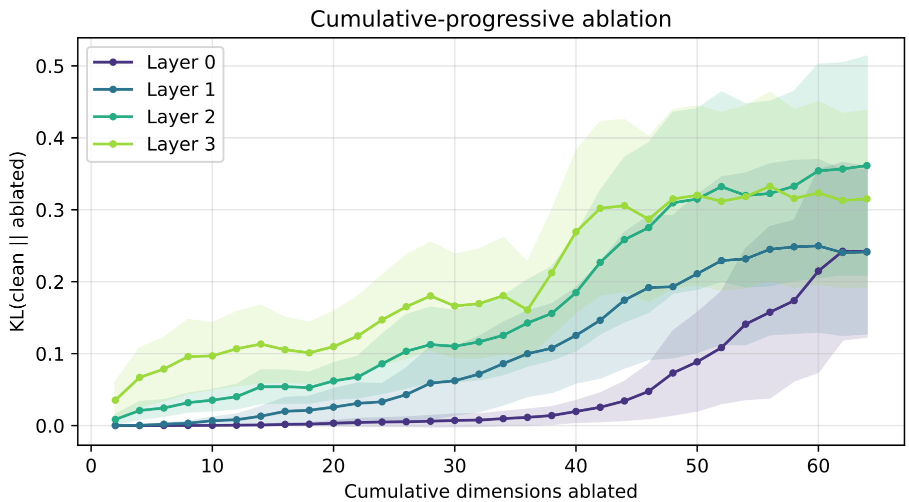


Figure 4: Cumulative-progressive ablation: KL divergence when subspaces $0, \ldots, k$ are jointly mean-ablated, as a function of the number of ablated dimensions $2(k+1)$. Shaded regions show $\pm 1$ standard deviation across sequences.

The per-iteration ablations (Figure 3) show each subspace's importance *in isolation*. A complementary question is: how does the causal impact accumulate as we ablate progressively more subspaces? Figure 4 answers this by jointly ablating subspaces 0 through $k$ for each $k = 0, 1, \ldots, 31$, covering 2 to 64 dimensions.

The cumulative-progressive curves reveal several features:

- **Layer 3** rises steeply in the first few iterations (driven by the dominant iteration-0 subspace), then continues to grow steadily as additional subspaces contribute incrementally. The final value (all 64 dims ablated) represents a complete erasure of all linear structure in the residual stream.

- **Layer 0** remains nearly flat across all 64 dimensions, confirming that belief-state-correlated directions at layer 0 are collectively as well as individually inert.

- At all layers, the curve is concave: the *marginal* contribution of each additional subspace diminishes. The first 2 dimensions account for the majority of the causal signal; subsequent subspaces add only small increments.

- The layer ordering is preserved throughout the entire range: layer 3 $>$ layer 2 $>$ layer 1 $>$ layer 0 at every value of $k$.

Figure 5 juxtaposes the two perspectives. The flat $R^2$ curves and the steeply declining KL bars make the dissociation visually clear: decodability does not imply causal relevance.

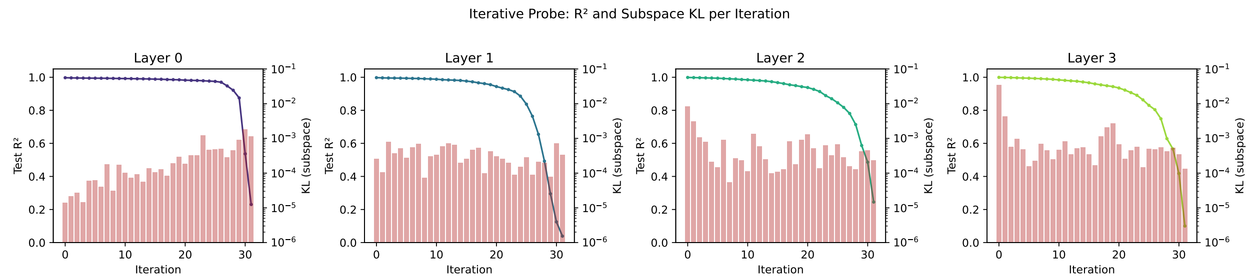


Figure 5: Dual-axis plots showing decodability ($R^2$, line) and causal importance of individual subspaces (KL, red bars) together for each layer. The contrast between flat $R^2$ and sharply decaying KL is most pronounced at layer 3.

## 5 Training Dynamics: Emergence of Redundant Copies

The redundancy documented above characterises the *final* model. A natural follow-up question is: does the multi-copy structure emerge during training, or is it already present at random initialisation?

To answer this, we ran the iterative probing algorithm (32 iterations, no $R^2$ cutoff) on four snapshots of the same architecture: the randomly initialised model (step 0), an early checkpoint (step 100), an intermediate checkpoint (step 2000), and the fully trained best model (∼46k steps). All four evaluations used the same 10,000 held-out sequences and belief states for comparability.

Figure **??** reveals that the redundant copy structure is **genuinely learned**, not an artefact of random initialisation. Several observations stand out:

1. **Random init baseline.** Even at initialisation, the belief state is partially decodable ($R^2 \approx 0.93$ at iteration 0). This is expected: random linear projections of a structured signal retain significant linear predictability, especially in a 64-dimensional space projecting a 2D target. However, $R^2$ decays *steadily* with each iteration—there is no plateau, and all layers are nearly indistinguishable.

2. **Early training (step 100).** The first-iteration $R^2$ increases slightly (∼0.93 → 0.94), and a mild layer ordering begins to appear, but the decay profile remains qualitatively similar to random init.

3. **Intermediate training (step 2000).** The first-iteration $R^2$ rises further to ∼0.96, and a shallow plateau begins to form over the first 10–15 iterations. Layer differentiation becomes more pronounced, with deeper layers showing faster decay after the plateau—the opposite of the best-model pattern.

4. **Fully trained (best model).** A qualitative phase transition: $R^2$ stays above 0.98 for ∼20 iterations, forming the extended plateau documented in Section 4. Layer 0 now retains the highest $R^2$ longest, while layer 3 degrades first—a reversal of the random-init ordering.

The contrast between random init and the trained model is most informative. At random init, the belief-state signal is "smeared" uniformly across all 64 dimensions by the random weight matrices; the steady $R^2$ decay simply reflects the decreasing number of available dimensions. Training reorganises the residual stream so that belief-state information is encoded with much higher fidelity and maintained across many more orthogonal projections.

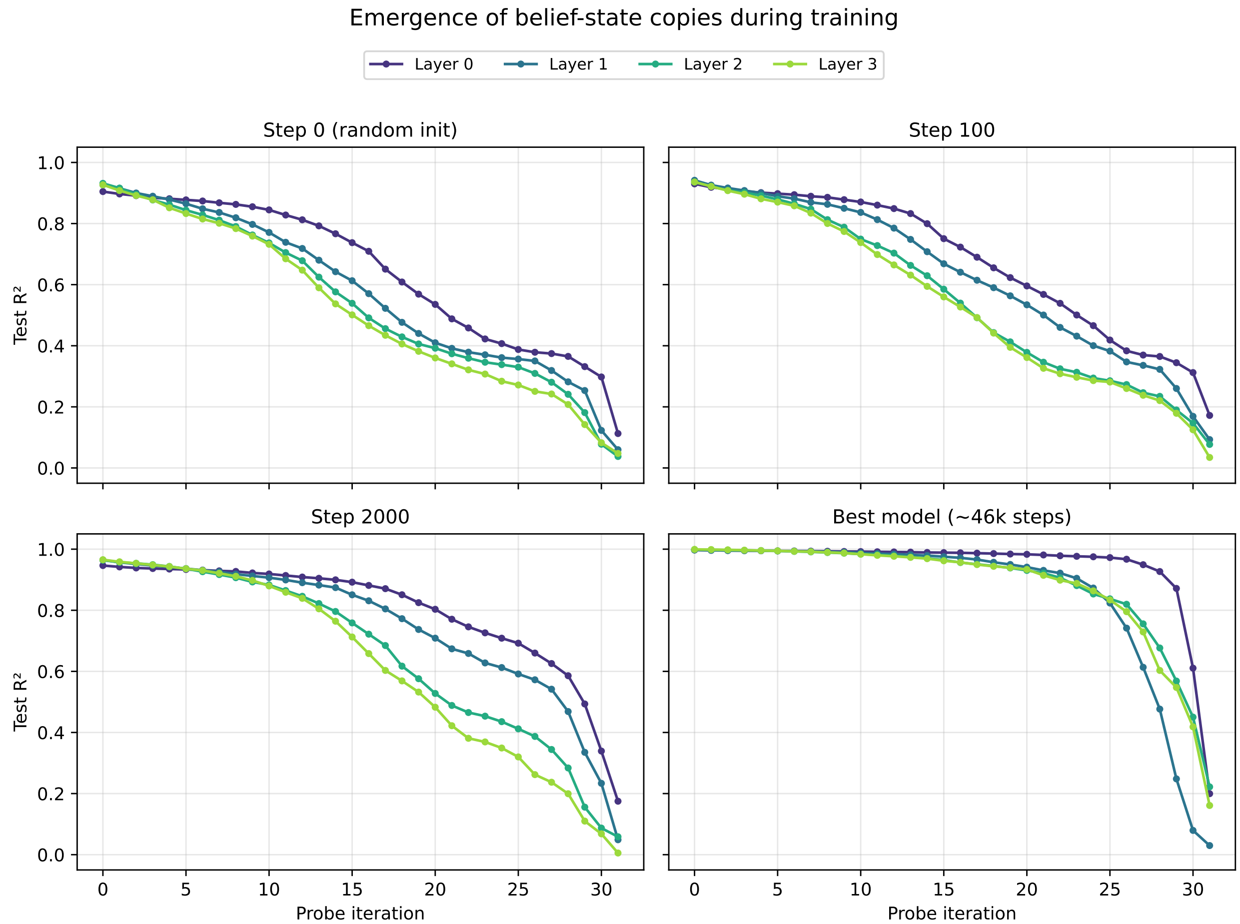


Figure 6: $R^2$ vs. probe iteration at four training stages. At random initialisation (top left), $R^2$ decays steadily from $\sim$0.93, with all layers behaving similarly. By step 100 (top right), the curves are slightly lifted but still show no plateau. At step 2000 (bottom left), the trained structure is emerging but the extended plateau is absent. The best model (bottom right) shows a qualitatively different regime: $R^2$ remains above 0.98 through $\sim$20 iterations before collapsing.

The layer ordering reversal is also notable. At random init, all layers behave identically (since each is just a different random projection). In the trained model, the layer hierarchy (layer 0 most robust, layer 3 degrading first in terms of redundancy) reflects the functional specialisation induced by training. Taken together with the ablation results of Section 4, this suggests that training concentrates the *causally important* belief-state signal into a small subspace at layer 3, while distributing redundant (but non-causal) copies more broadly at earlier layers.

# 6 Discussion

## 6.1 The Nature of Redundancy

The fact that the belief state is linearly readable from over 25 independent subspaces spanning most of the 64-dimensional residual stream raises the question: is this deliberate or incidental?

In a linear model, the output logits are computed as logit $= W_U\, a + b_U$, where $W_U \in \mathbb{R}^{3\times 64}$

is the unembedding matrix and $a$ is the final residual stream. The unembedding only "reads" the projection of $a$ onto the row space of $W_U$, which is at most 3-dimensional (for a 3-token vocabulary). In a transformer with nonlinear MLP layers and attention, the situation is more complex: intermediate layers can route information through different subspaces, so multiple copies at an earlier layer may serve different downstream computations.

However, the ablation results suggest that even at the *final* layer, only the first iteration's subspace carries substantial causal weight. The remaining copies appear to be **representational byproducts**—linear structure that arises naturally from the network's computation but is not directly utilised for the output.

One plausible mechanism is that each attention head and MLP layer writes a belief-state-correlated component into the residual stream as part of its computation, and these components accumulate across layers. Since the residual stream is a sum of all layer outputs, any layer that computes a function of the belief state will leave a linearly decodable trace, even if that trace is not read by subsequent layers.

## 6.2 Layer Hierarchy and the Residual Stream

The exponential growth of causal importance across layers ($0.003 \to 0.028 \to 0.058 \to 0.112$) aligns with the standard picture of transformer computation: early layers build up representations that are progressively refined, and the final layer's output is what directly determines predictions.

The near-zero causal importance of layer 0's belief-state subspace is particularly striking. The model encodes the belief state with $R^2 = 0.997$ at layer 0, yet ablating all 20 belief-state-correlated dimensions barely affects the output. This means the subsequent layers do not read the belief state from the directions where a linear probe finds it at layer 0. They may instead reconstruct it through attention and MLP operations, or read it from entirely different (nonlinear) features of the early residual stream.

## 6.3 Decodability $\neq$ Causal Use

This experiment provides a concrete demonstration of a well-known but often underappreciated point in mechanistic interpretability: **the existence of a linear probe does not establish causal relevance**. All 40 subspaces across 4 layers and 10 iterations have $R^2 > 0.98$, yet their KL divergences span nearly four orders of magnitude (from $10^{-5}$ to $10^{-2}$).

The iterative probing methodology makes this dissociation quantitative. It allows us to say not just "the belief state is decodable" but "the belief state is decodable from $N$ independent subspaces, of which only the first at layer $L$ is causally important."

## 6.4 Comparison to Superposition

In the superposition literature, networks are known to store more features than they have dimensions by encoding features in nearly-orthogonal directions. The pattern here is different: the same 2D feature (the belief state) is stored in many exactly-orthogonal copies. This is closer to **redundancy** than superposition.

One interpretation is that this redundancy is an artefact of overparameterisation: the network has 64 dimensions but only needs $\sim$2 to represent the belief state, so the gradient-based optimisation naturally distributes belief-state-correlated structure across many directions. The ablation results suggest that the model has not learned to "clean up" this redundancy—it uses a specific 2D subspace and ignores the rest.

## 7 Implementation Notes

The implementation is split across two files:

- src/iterative_probe.py: Pure numpy/sklearn logic for the iterative probing algorithm. Contains the dataclasses IterationResult and IterativeProbeResult, and the functions extract_subspace(), project_out_subspace(), iterative_probe_single_layer(), and iterative_probe
- src/run_iterative_probe.py: CLI pipeline that orchestrates data generation, activation extraction (reusing src/activation_extraction.py), iterative probing (Phase 1), and counterfactual ablation (Phase 2). The ablation helpers ablate_subspace() and compute_counterfactual() are defined here.

The CLI supports --skip-counterfactual for running Phase 1 alone, and --force to re-run over existing results. Output is saved to {model_dir}/iterative_probe_results/ as three files: iterative_metrics.csv, counterfactual_metrics.csv, and subspaces.npz.

## 8 Conclusion

Iterative probing reveals that a 4-layer, 64-dimensional transformer trained on the Mess3 HMM stores the 2D belief state across nearly all of its residual stream dimensions: $R^2$ remains above 0.93 through 25 successive projections (50 dims removed) before collapsing only as the space is exhausted. Despite this pervasive redundancy, counterfactual ablation—both per-subspace and cumulative-progressive—shows that only the first subspace at the final layer carries substantial causal weight for next-token prediction. The cumulative-progressive curves confirm that marginal causal importance diminishes rapidly: the first 2 dimensions at layer 3 account for the lion's share of the KL shift, with subsequent subspaces contributing only incrementally.

This dissociation between decodability and causal importance has methodological implications: probe $R^2$ alone is insufficient for understanding which representations a network actually relies upon. The iterative probing framework, augmented with cumulative-progressive ablation, provides a structured way to quantify both the degree of redundancy and the causal hierarchy among redundant copies.

## References


[1] S. Shai, S. Vardi, I. Arad, and N. Razin, "Transformers Represent Belief State Geometry in their Residual Stream," *arXiv preprint arXiv:2405.15943*, 2024.